

Buon Kiong Lau

Characteristic Modes: Into the Mainstream and the Path Beyond

Around 15 years ago, I started to collaborate with Sony (Sony Ericsson at the time) on designing compact multiantennas for multiple-input, multiple-output (MIMO) terminals. It wasn't long before I realized that the mutual coupling between two antenna elements was extreme at the low cellular band (below 1 GHz), even though they were placed as far away from each other as possible on the terminal chassis. This coupling was far higher than what could be expected from two stand-alone antennas separated by the same distance.

This phenomenon later led me to the interesting works of the late Prof. Pertti Vainikainen and his group on terminal antennas at the low band (e.g., [1]). Basically, the antenna elements become "coupling elements" for the chassis, which serves as the main radiator. Therefore, if two elements are used, and they both rely on the chassis for radiation, they will be tightly coupled. Inspired by the early work of Prof. Werner Schroeder in applying characteristic mode analysis (CMA) to analyze and utilize the effect of the terminal chassis [2], my group started to look into MIMO antenna design for mobile terminals using CMA.

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The results (in terms of both gained insights and achieved performances) were well beyond our expectations! But we were surprised that this tool was scarcely known in the antenna community. This is hardly an isolated story, but one of many on how CMA has been rediscovered by antenna researchers during recent decades.

By 2014, after a special session organized around this topic at the IEEE International Symposium on Antennas and Propagation (AP-S) and the U.S. National Committee International Union of Radio Science (URSI) Meeting (AP-S/URSI) in Memphis, Tennessee, a few like-minded colleagues and I called an ad hoc meeting for those who were active in the topic, and together we decided to set up the Characteristic Modes Special Interest Group (CM-SIG) for the purpose of sharing our exciting findings and coordinating our efforts relating to this topic. Among our success stories, we have successfully organized a special issue in *IEEE Transactions on Antennas and Propagation* [3], a European level training school under the framework of the European School of Antennas, and regular special/convened sessions at the two largest antenna conferences, i.e., AP-S/URSI and the European Conference on Antennas and Propagation.

As the years went by, it became clear to us that the time is ripe for a review series on the topic. What better place can we consider publishing such a work than in *IEEE Antennas and Propagation Magazine*? Our ambition is to present how the topic has been developing, especially over the past few years, during which hundreds of articles have been published on the subject. The significant academic output has been occurring hand in hand with the increasing adoption of CMA in commercial electromagnetic software packages, which also facilitates the applications of CMA in antenna and electromagnetic engineering. We would like to offer the community a concise overview, capturing as much as possible the main advances expedited on, and by, CMA as well as the lessons learned.

To this end, this review series consists of five articles focusing on the theory [4], [5] and main application areas [6]–[8] of CMA. In particular, Lau et al. [4]

provide an overall introduction and also briefly address some emerging applications, including scattering and the use of dielectric/magnetic materials. Capek and Schab's article [5] deals with the computational aspects of CMA, critical for its successful application, which requires characteristic modes (CMs) from a continuous theoretical framework to be transformed to a discrete representation compatible with numerical methods.

With earlier works focusing on the utilization of CMA for the analysis and enhancement of antenna element designs, Adams et al. [6] spearhead the application-focused articles in the review series. This is followed by the convenient treatment of platform-integrated antennas using CMA in Li et al.'s article [7]. As a fitting end to the review series, Manteuffel et al. [8] present an overview of the application of CMA in the design of arrays and metasurface antennas, a topic that is very timely because of the great interest in applying high-gain antennas to enable 5G/6G systems. To ensure high-quality work, senior authors active in relevant topics were tasked with leading the writing of these articles with as much support as possible from coauthors from other institutions and the CM community represented by the CM-SIG.

I would like to take this opportunity to thank the authors and reviewers of all five articles for their great efforts in making this review series a reality—and with the articles reviewed, revised, and accepted ahead of schedule! Sincere thanks go to the CM-SIG community, who strongly supported this initiative, even though we could not involve everyone in the actual writing. Finally, the extraordinary support and encouragement from *IEEE Antennas and Propagation Magazine* Editor-in-Chief Prof. Francesco P. Andriulli and Editorial Assistant Christina Tang-Bernas have made the journey of organizing this review series a pleasant one. It is my sincere hope that all of you will enjoy reading these articles.

AUTHOR INFORMATION

Buon Kiong Lau (buon_kiong.lau@eit.lth.se) is with the Department of Electrical and Information Technology, Lund University, Lund, 221 00, Sweden.

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